

TRACE MAIL AI

SIH26106 — AI-Powered Email Threat Detection, Geolocation & Forensic Intelligence Platform

Prototype Specification & 2-Day Hackathon Build Plan

Core positioning: TraceMail AI does not stop at detecting a suspicious email. It explains the evidence, reconstructs observed source infrastructure, connects related incidents into campaigns, supports investigation, and generates an evidence-backed forensic report.

Core workflow

Detect → Explain → Trace → Correlate → Investigate → Report

Prototype objective

Build a convincing, lightweight forensic investigation platform rather than a generic phishing classifier. The prototype should demonstrate one realistic BEC/phishing case from EML upload through campaign correlation and report generation.

1. Problem & Solution

Problem statement focus

SIH26106 asks for a platform that can detect phishing, spoofed, impersonated and fraudulent emails; analyze headers, metadata, body, URLs and attachments; inspect SPF/DKIM/DMARC and relay paths; enrich IP/domain infrastructure with geolocation and threat intelligence; correlate related indicators and campaigns; and provide a searchable forensic investigation workflow with evidence preservation and reporting.

Proposed solution

TraceMail AI combines deterministic email forensics, threat-intelligence enrichment and AI-assisted reasoning to turn an isolated suspicious email into a connected investigation.

Target users

- SOC and cybersecurity analysts
- Incident response and CERT teams
- Enterprise or institutional security teams
- Fraud and BEC investigators
- Cybercrime / digital-forensics investigators

2. What Makes the Prototype Different

Individual capabilities such as phishing detection, SPF/DKIM/DMARC checks, IP reputation, geolocation and AI email analysis already exist in various tools. Therefore, do not claim that every feature is individually new. The differentiation is the integrated evidence-driven investigation workflow.

Differentiator	What it does	Why it matters
AI Attack Hypothesis	Compares BEC, credential phishing, spoofing, malware delivery and other possible types of phishing	Moves beyond binary phishing/not-phishing classification.
Evidence Reasoner	Combines verified technical signals with AI/NLP findings	Makes the results explainable and reduces unsupported AI claims
Campaign DNA	Finds relationships across suspicious emails using domains, URLs, IP addresses, etc.	Turns single IP addresses into a campaign-level intelligence.
Evidence / Infrastructure Graph	Connects email, domains, URLs, IPs, ASN/hosting, hashes and related data	Shows investigated cases relationships rather than isolated IOC lists
Investigator Copilot	Answers evidence-specific questions and recommends next steps	Provides analyst assistance without becoming a generic chatbot

3. Prototype Modules

3.1 Evidence Intake & Case Creation

- Upload a raw .eml file or paste raw email headers.
- Create a unique Case ID.
- Preserve original evidence metadata.
- Calculate SHA-256 hash of the original EML.
- Store upload timestamp and basic audit information.
- Optional: upload an attachment for static analysis.

3.2 Email Forensic Analyzer

- Extract From, To, Reply-To, Return-Path, Subject, Date and Message-ID.
- Parse all Received headers into a relay chain.

- Extract Authentication-Results.
- Evaluate or display SPF, DKIM and DMARC results where technically available.
- Extract URLs, domains, IP addresses and attachment names/hashes.
- Flag suspicious mismatches and routing anomalies.
- Show both the raw evidence and a human-readable explanation.

3.3 AI Attack Hypothesis Engine

The AI should compare plausible attack scenarios instead of returning only one generic phishing score.

Hypothesis	Example signals
BEC / Executive impersonation	Executive identity, urgency, payment request, Reply-To mismatch, suspicious domain.
Credential phishing	Login lure, suspicious URL, brand impersonation, credential-collection language.
Spoofing / impersonation	Display-name deception, domain mismatch, authentication failures.
Malware delivery	Suspicious attachment, executable/script indicators, malicious URL or delivery pattern.
Generic spam / low risk	Weak malicious indicators and no strong attack evidence.

3.4 Evidence Reasoner

The system should combine deterministic evidence, threat-intelligence results and AI/NLP signals. The LLM should reason over structured evidence instead of inventing technical facts.

Example output: SPF failure + Reply-To mismatch + lookalike domain + suspicious URL + urgent payment request → high likelihood of BEC.

3.5 Infrastructure Intelligence

- IP geolocation: country/region/city where available.
- ASN, ISP/hosting and network information.
- Domain registration/RDAP information where available.
- DNS and MX information.
- IP/domain/URL reputation and threat-intelligence results.
- Indicators for hosting, VPN/proxy or other infrastructure characteristics where reliable data exists.

Forensic wording: Say “observed/probable source infrastructure” rather than “the attacker’s exact location.” IP geolocation identifies infrastructure, not necessarily the human attacker.

3.6 Campaign DNA

- Compare the current case with stored suspicious cases.
- Match shared domains, URLs, IPs, ASN/hosting, attachment hashes and other IOCs.
- Use AI/semantic similarity only as an additional signal.
- Produce a campaign similarity score and show the reasons for the match.
- Example: “Potentially related to Campaign CMP-0042 — 7 cases, 87% similarity.”

3.7 Evidence / Infrastructure Graph

Recommended nodes: Email, Case, Sender, Reply-To, Domain, URL, IP, ASN/Hosting, Attachment Hash and Campaign.

Recommended edges: sent-from, replies-to, contains, resolves-to, hosted-by, shares-infrastructure-with, related-to.

The graph should be clickable so selecting a node reveals its evidence and related cases.

3.8 Investigation Timeline

- Display observed relay and delivery events in chronological order.
- Separate OBSERVED facts from INFERRED conclusions and UNKNOWN information.
- Avoid presenting assumptions as confirmed attacker actions.

3.9 Investigator Copilot

Use investigation-specific actions rather than a generic chatbot:

- Why is this email risky?
- What evidence supports BEC?
- Show related cases.
- Explain the source infrastructure.
- What should I investigate next?
- Generate forensic summary.

3.10 Case Management & Report

- Case list with status, severity and attack type.
- Case tabs: Overview, Evidence, Email Analysis, Threat Intel, Graph, Campaign, Timeline, AI Findings and Report.
- Generate a PDF forensic report containing evidence, findings, IOCs, timeline, confidence, limitations and recommendations.
- Include SHA-256 evidence hash and audit information.

4. Recommended Dashboard

The dashboard should immediately communicate investigation status.

Area	Display
Top summary	Risk score, attack hypothesis, confidence, number of related cases.
Authentication	SPF, DKIM, DMARC, Reply-To/Return-Path indicators.
Threat indicators	Suspicious URLs, domains, IPs, attachments and reputation findings.
Infrastructure	Observed relay/source infrastructure, ASN/hosting and geolocation.
Graph	Interactive Email → Domain/URL → IP/ASN → Campaign relationships.
AI findings	Attack hypothesis and evidence-backed explanation.
Timeline	Observed/inferred investigation timeline.
Actions	Find campaign, ask Copilot, export report.

Suggested navigation

Dashboard | New Investigation | Email Forensics | Threat Intelligence | AI Investigation | Campaigns | Investigation Graph | Timeline | Cases | Reports

Suggested case view

Case TM-00124 — BEC — HIGH — OPEN

Overview | Evidence | Email Analysis | Threat Intel | Graph | Campaign | Timeline | AI Findings | Report

5. Technical Architecture

Keep the architecture practical for a 2-day student hackathon. Use deterministic modules for technical facts and an API-based LLM for reasoning.

Data flow

EML Upload → Email Parser → Forensic Engine + AI Engine → Risk/Evidence Correlation → Threat Intelligence → Campaign/Graph Engine → Investigator Dashboard → Forensic Report

Layer	Recommended technology	Purpose
Frontend	HTML, CSS, JavaScript	Lightweight responsive dashboard.
Optional frontend upgrade	React	Use only if a team member is comfortable with it.
Backend	Python + Flask or FastAPI	API endpoints, orchestration and business logic.
Email parsing	Python email/parser libraries	Read EML, headers, body, MIME parts and attachments.
Database	SQLite	Cases, IOCs, evidence metadata, campaign links and audit data.
AI/LLM	API-based LLM	NLP, attack hypotheses, evidence reasoning, campaign similarity and Copilot.
Graph	Cytoscape.js or vis-network	Interactive evidence/infrastructure graph.
Maps	Leaflet + OpenStreetMap	Infrastructure geolocation visualization.
Charts	Chart.js	Risk and investigation metrics.
Threat intelligence	Selected reputation/domain/IP APIs	Enrichment of observed IOCs.
Report	Python PDF generation	Forensic report export.
Security/integrity	SHA-256 + audit log	Evidence integrity and traceability.

AI vs deterministic responsibilities

Deterministic / verified	AI-assisted
EML parsing	Phishing/BEC language analysis
Header and Received-chain parsing	Semantic impersonation detection
SPF/DKIM/DMARC results	Attack hypothesis generation
URL/IP/domain extraction	Evidence reasoning over structured findings
DNS/RDAP/IP/ASN enrichment	Campaign semantic similarity
Attachment hashing	Investigator Copilot responses
SHA-256 and audit log	Human-readable forensic narrative

6. Data Model for the Prototype

Keep the database small. Suggested core tables:

Table	Important fields
cases	case_id, title, status, severity, attack_type, confidence, created_at
evidence	id, case_id, filename, sha256, size, uploaded_at
emails	id, case_id, from_addr, reply_to, return_path, subject, date, message_id
headers	id, email_id, header_name, header_value, hop_order, trust_status
iocs	id, case_id, type, value, reputation, source
infrastructure	id, ioc_id, country, city, isp, asn, hosting, notes
campaigns	campaign_id, name, similarity, description
case_campaigns	case_id, campaign_id, match_reason
ai_findings	case_id, hypothesis, score, evidence, limitations
audit_log	id, case_id, action, timestamp, actor

7. AI Output Contract

Ask the LLM to return structured JSON so the frontend can reliably display results. Example fields:

attack_hypotheses: [{type, score, reasons}]

evidence_summary: [verified evidence IDs]

contradictions: [signals that conflict]

recommended_actions: [next investigation steps]

confidence: low/medium/high

limitations: [uncertainties]

Important: The AI should receive structured facts from the forensic engine and threat-intelligence layer. Do not ask it to invent SPF, DKIM, DNS, geolocation or reputation results.

8. Signature Demo Scenario

Use one realistic BEC case for the entire presentation. Avoid switching between unrelated sample emails.

Scenario: An employee receives an urgent payment or bank-detail-change request that appears to come from a senior executive.

- Upload suspicious .eml.

- Show parsed From, Reply-To, Return-Path and Received chain.
- Show SPF/DKIM/DMARC results and mismatches.
- Extract suspicious URL/domain and enrich infrastructure.
- Run AI Attack Hypothesis: BEC / executive impersonation becomes the leading hypothesis.
- Open Evidence Reasoner and show the technical + AI evidence behind the conclusion.
- Show observed infrastructure on map and in the graph.
- Run Campaign DNA and connect the case to previous suspicious emails.
- Ask Investigator Copilot why the campaign match is strong and what to investigate next.
- Generate the forensic report.

Recommended final demo statement

“TraceMail AI does not stop at detecting the phishing email. It connects the evidence to explain the attack, trace the observed infrastructure, identify related campaigns and support the investigator with an auditable forensic report.”

9. 2-Day Implementation Plan

Time	Build priority	Deliverable
Day 1 — Block 1	Project skeleton + database + EML upload	Working case creation and evidence storage.
Day 1 — Block 2	Email parser + header/URL/IP extraction	Forensic evidence view.
Day 1 — Block 3	SPF/DKIM/DMARC + basic risk engine	Initial threat assessment.
Day 1 — Block 4	LLM Attack Hypothesis + Evidence Reasoner	Explainable AI findings.
Day 2 — Block 1	Threat-intel enrichment + infrastructure view	IP/domain intelligence.
Day 2 — Block 2	Campaign DNA + graph	Main uniqueness demonstration.
Day 2 — Block 3	Timeline + Investigator Copilot	Investigation workflow.
Day 2 — Block 4	Report + UI polish + demo rehearsal	Judge-ready prototype.

Priority rule

Must work: EML parsing → forensic evidence → AI hypothesis → infrastructure enrichment → campaign correlation → graph → report.

Can be simplified: advanced ML training, live mailbox monitoring, large-scale SIEM integration, complex authentication, or many external APIs.

10. Security, Privacy & Forensic Safeguards

- Do not claim exact human attribution from IP geolocation.
- Mark evidence as observed, inferred or unknown.
- Keep original EML unchanged and calculate its SHA-256 hash.
- Maintain basic case/audit logs.
- Mask unnecessary personal information in the UI where possible.
- Use role-based access if time permits.
- Clearly label demo/mock threat-intelligence data when an external API is unavailable.
- Store API keys in environment variables; never hard-code secrets.

11. What to Avoid

- Do not build a generic AI chatbot as the main feature.
- Do not claim 100% accurate phishing detection.
- Do not claim an IP address proves the attacker's physical location.
- Do not add blockchain unless it has a specific evidence-integrity purpose.
- Do not train a large custom AI model for a 2-day prototype.
- Do not add unrelated AI features just to increase the feature count.
- Do not depend on a large number of APIs for the core demo.

12. Final Feature Checklist

Priority	Feature
P0	EML upload and case creation
P0	Header / Received-chain forensic parsing
P0	SPF / DKIM / DMARC analysis
P0	URL, IP, domain and attachment extraction
P0	Risk and severity assessment
P0	AI Attack Hypothesis Engine
P0	Evidence Reasoner
P1	IP/domain/URL threat intelligence
P1	Infrastructure geolocation + ASN/hosting
P1	Campaign DNA / cross-email correlation
P1	Interactive evidence/infrastructure graph
P1	Investigation timeline
P1	Investigator Copilot
P1	Case management
P1	SHA-256 evidence integrity + audit log
P1	PDF forensic report
P2	Advanced role-based access / retention controls
P2	More external integrations and advanced ML

13. Judge-Ready Innovation Statement

Problem: Existing security tools can expose individual indicators, but investigators often need to manually correlate email headers, authentication results, URLs, domains, IP infrastructure, reputation and related incidents.

Our approach: TraceMail AI integrates these evidence sources into a single investigation workflow and adds AI-assisted attack hypotheses, explainable evidence reasoning, campaign correlation and an investigator copilot.

Innovation: The platform is designed around **connected evidence and campaign-level investigation**, not merely single-email classification.

One-line innovation

14. Final Prototype Blueprint

Frontend: Dashboard, case pages, evidence panels, graph, map, timeline, Copilot and report actions.

Backend: Flask/FastAPI orchestration and case APIs.

Forensic engine: EML/MIME parsing, headers, Received chain, SPF/DKIM/DMARC, URLs, IPs, attachments and hashes.

Threat-intel engine: selected IP/domain/URL enrichment APIs with cached/mock fallback for demo reliability.

AI engine: structured attack hypotheses, evidence reasoning, campaign similarity and investigation assistance.

Correlation engine: IOC relationships and Campaign DNA.

Graph engine: interactive investigation graph.

Evidence layer: SHA-256, audit log, observed/inferred/unknown labels.

Output: dashboard + connected investigation + forensic PDF report.

Bottom line

For a 2-day SIH prototype, prioritize depth over feature count. A smaller system that successfully demonstrates **EML forensics** → **explainable AI** → **infrastructure tracing** → **Campaign DNA** → **graph** → **investigator workflow** → **report** will be stronger than a large collection of disconnected AI features.